



MANIPAL UNIVERSITY JAIPUR
School of Computer Science and Engineering
DEPARTMENT OF DATA SCIENCE AND ENGINEERING

Course Handout: B.Tech DSE 7th Semester

Computer Vision | DS 4150 | 3 Credits | 3 0 0 3

Session: July 25 – Nov 25 | Faculty: Dr. Aishwarya Mishra

A. Introduction: This course on Computer Vision provides a comprehensive foundation in image processing techniques, starting with the fundamental principles of image formation and digitization. Students will explore key operations such as pixel connectivity, image enhancement, and feature extraction, leading to advanced topics in image segmentation and object recognition. By the end of the course, participants will be equipped with the skills to apply computer vision techniques across a range of applications, including biometrics, medical imaging, and augmented reality.

B. Course Outcomes: At the end of the course, students will be able to:

[DS 4150.1] Remember key concepts and terminologies in image processing system.

[DS 4150.2] Explain pixel connectivity concepts and basic image processing operations.

[DS 4150.3] Apply various image pre-processing techniques to enhance digital images.

[DS 4150.4] Analyze digital images and extract features using different feature extraction techniques.

[DS 4150.5] Evaluate and justify the use of specific segmentation techniques for different image processing tasks.

[DS 4150.6] Implement solutions for real-world applications using image processing and computer vision techniques.

C. PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES

PROGRAM OUTCOMES

[PO.1]. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering **specialization** to the solution of complex engineering problems.

[PO.2]. **Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

[PO.3]. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

[PO.4]. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

[PO.5]. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

[PO.6]. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

[PO.7]. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO.8]. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices.

[PO.9]. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

[PO.10]. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

[PO.11]. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

[PO.12]. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

[PSO.1] Understand the role of mathematics, statistics, and AI techniques in the field of data science & engineering.

[PSO.2] Apply the acquired knowledge and expertise to perform data analytics tasks for multidimensional datasets.

[PSO.3] Develop effective and scalable industrial solutions for real world socio-economic problems using data analytics tools and techniques.

D. Assessment Plan:

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid Term Exam (Close Book)	30
	In class Quizzes and Assignments, Activity (Accumulated and Averaged)	30
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100
Attendance (Formative)	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking up the End Semester examination. The allowance of 25% includes all types of leaves including medical leaves.	
Make up Assignments (Formative)	Students who miss a class will have to report to the teacher about the absence. A makeup assignment on the topic taught on the day of absence will be given which has to be submitted within a week from the date of absence. No extensions will be given on this. The attendance for that day of absence will be marked blank, so that the student is not accounted for absence. These assignments are limited to a maximum of 5 throughout the entire semester.	
Homework/ Home Assignment/ Activity Assignment (Formative)	There are situations where a student may have to work in home, especially before a flipped classroom. Although these works are not graded with marks. However, a student is expected to participate and perform these assignments with full zeal since the activity/ flipped classroom participation by a student will be assessed and marks will be awarded.	

E. SYLLABUS

DS4150: Computer Vision [3 1 0 4]

Introduction to image processing: Overview and significance of image processing, Components of an image processing system, Image formation principles, Digitization of images, Continuous and discrete image representations,

Basic image processing operations: Neighbours of pixel, Adjacency and connectivity, Regions and boundaries, Distance measures, Arithmetic operations, Logical operations, Geometrical operations, Convolution and correlation operations, **Image pre-processing techniques and feature extraction:** Noise reduction and filtering, Smoothing and sharpening, Histogram equalization, Image enhancement techniques, Point detection, Line and edge detection, Colour, texture, shape and structure features, Spatial and frequency domain features, Corner detection, Hough transform, **Image segmentation:** Boundary detection techniques, Point, line, and edge detection, Edge linking and local processing, regional processing, Hough Transform for segmentation, Thresholding techniques: Iterative, Otsu's method, Multivariable, Region-based segmentation, Watershed algorithm, Motion-based segmentation, **Applications and Fundamentals of Image Processing and Computer Vision:** Applications of image processing- Document image analysis, Biometrics, Object recognition and tracking, medical image analysis, Content-based image retrieval, Video data processing, Multimedia applications, Virtual reality and augmented reality, Fundamentals of computer vision-What is computer vision? Low-level, Mid-level and High-level computer vision tasks, Overview of computer vision applications. **Techniques and Representations in Computer Vision:** Low-level vision techniques- Edges, Contours, Textures, Shapes, and Colours, Motion Detection, Optical Flow, and Tracking, Local Features and Invariance, Bag-of-words models, Fisher vector, Mid-level representations-Object parts and attributes, Embedding and representing objects, Understanding and organizing parts and structures

REFERENCE BOOKS

1. David A Forsyth & Jean ponce “Computer vision – A modern approach”, Prentice Hall, Pearson Education India; Edition: Second, ISBN-13: 978-9332550117, 2015.
2. R. C. Gonzalez, R. E. Woods. “Digital Image Processing” Pearson, Inc., Edition-Fourth, ISBN. 978-0131687288, 2017.
3. A. K. Jain. “Fundamentals of Digital Image Processing” Prentice-Hall, Pearson; Edition: First, ISBN-13: 978-0133361650, 1994.
4. David A Forsyth & Jean Ponce, “Computer vision – A modern Approach”, Prentice Hall ,2002.
5. Bernd Jahne and Horst Hau Becker, “Computer vision and Applications” Academic press, 2000.

F. Lecture Plan:

Lecture No.	Major Topics	Topics	Mode of Delivery	Corresponding CO	Mode of Assessing the Outcome
1	Introduction to image processing	Overview and significance of image processing, Components of an image processing system	Lecture	DS 4150.1	Class Quiz Mid Term
2	Introduction to image processing	Image formation principles	Lecture	DS 4150.1	Class Quiz Mid Term
3	Introduction to image processing	Digitization of images, Continuous and discrete image representations	Lecture	DS 4150.1	Class Quiz Mid Term
4	Introduction to image processing	Digitization of images, Continuous and discrete image representations	Lecture	DS 4150.1	Class Quiz Mid Term
5	Introduction to image processing	Digitization of images, Continuous and discrete image representations	Lecture	DS 4150.1	Class Quiz Mid Term
6	Introduction to image processing	Digitization of images, Continuous and discrete image representations	Lecture	DS 4150.1	Class Quiz Mid Term
7	Basic image processing operations	Neighbours of pixel, Adjacency and connectivity	Lecture	DS 4150.2	Class Quiz Mid Term
8	Basic image processing operations	Regions and boundaries, Distance measures	Lecture	DS 4150.2	Class Quiz Mid Term

9	Basic image processing operations	Arithmetic operations, Logical operations	Lecture	DS 4150.2	Class Quiz Mid Term
10	Basic image processing operations	Geometrical operations	Lecture	DS 4150.2	Class Quiz Mid Term
11	Basic image processing operations	Convolution and correlation operations	Lecture	DS 4150.2	Class Quiz Mid Term
12	Image pre-processing techniques and feature extraction	Noise reduction and filtering	Lecture	DS 4150.3	Class Quiz Mid Term
13	Image pre-processing techniques and feature extraction	Smoothing and sharpening	Lecture	DS 4150.3	Class Quiz Mid Term
14	Image pre-processing techniques and feature extraction	Histogram equalization	Lecture	DS 4150.3	Class Quiz Mid Term
15	Image pre-processing techniques and feature extraction	Image enhancement techniques	Lecture	DS 4150.3	Class Quiz Mid Term
16	Image pre-processing techniques and feature extraction	Point, line and edge detection	Lecture	DS 4150.3	Class Quiz Mid Term
17	Image pre-processing techniques and feature extraction	Colour, texture, shape and structure features	Lecture	DS 4150.3	Class Quiz Mid Term
18	Image pre-processing techniques and feature extraction	Spatial and frequency domain features	Lecture	DS 4150.3	Class Quiz Mid Term
19	Image pre-processing techniques and feature extraction	Spatial and frequency domain features	Lecture	DS 4150.3	Class Quiz Mid Term
20	Image pre-processing techniques and feature extraction	Spatial and frequency domain features	Lecture	DS 4150.3	Class Quiz Mid Term
21	Image pre-processing techniques and feature extraction	Spatial and frequency domain features	Lecture	DS 4150.3	Class Quiz Mid Term
22	Image pre-processing techniques and feature extraction	Spatial and frequency domain features	Lecture	DS 4150.3	Class Quiz Mid Term
23	Image pre-processing techniques and feature extraction	Corner detection	Lecture	DS 4150.3	Class Quiz Mid Term
24	Image pre-processing techniques and feature extraction	Hough transform	Lecture	DS 4150.3	Class Quiz Mid Term
25	Image segmentation	Boundary detection techniques, Point, line, and edge detection	Lecture	DS 4150.4	Class Quiz Mid Term
26	Image segmentation	Edge linking and local processing, Regional processing	Lecture	DS 4150.4	Class Quiz Mid Term
27	Image segmentation	Hough Transform for segmentation	Lecture	DS 4150.4	Class Quiz Mid Term
28	Image segmentation	Thresholding techniques: Iterative, Otsu's method, Multivariable	Lecture	DS 4150.4	Class Quiz Mid Term
29	Image segmentation	Region-based segmentation, Watershed algorithm,	Lecture	DS 4150.4	Class Quiz Mid Term

		Motion-based segmentation			
Md Term Exam					
30	Applications and Fundamentals of Image Processing and Computer Vision	Applications of image processing in document analysis and biometrics	Lecture	DS 4150.5	Class Quiz End Term
31	Applications and Fundamentals of Image Processing and Computer Vision	Applications of image processing in object recognition and tracking and medical image analysis	Lecture	DS 4150.5	Class Quiz End Term
32	Applications and Fundamentals of Image Processing and Computer Vision	Application of image processing in content-based image retrieval and video data processing	Lecture	DS 4150.5	Class Quiz End Term
33	Applications and Fundamentals of Image Processing and Computer Vision	Application of image processing in multimedia applications and virtual reality, augmented reality	Lecture	DS 4150.5	Class Quiz End Term
34	Applications and Fundamentals of Image Processing and Computer Vision	Application of image processing in multimedia applications and virtual reality, augmented reality	Lecture	DS 4150.5	Class Quiz End Term
35	Applications and Fundamentals of Image Processing and Computer Vision	Application of image processing in multimedia applications and virtual reality, augmented reality	Lecture	DS 4150.5	Class Quiz End Term
36	Applications and Fundamentals of Image Processing and Computer Vision	Application of image processing in multimedia applications and virtual reality, augmented reality	Lecture	DS 4150.5	Class Quiz End Term
37	Applications and Fundamentals of Image Processing and Computer Vision	Application of image processing in multimedia applications and virtual reality, augmented reality	Lecture	DS 4150.5	Class Quiz End Term
38	Applications and Fundamentals of Image Processing and Computer Vision	Fundamentals of Computer Vision	Lecture	DS 4150.5	Class Quiz End Term
39	Applications and Fundamentals of Image Processing and Computer Vision	Low-Level, Mid-Level, and High-Level Computer Vision Task	Lecture	DS 4150.5	Class Quiz End Term
40	Applications and Fundamentals of Image Processing and Computer Vision	Overview of Computer Vision Applications	Lecture	DS 4150.5	Class Quiz End Term
41	Techniques and Representations in Computer Vision	Edges, Contours, Textures, Shapes, and Colours	Lecture	DS 4150.6	Class Quiz End Term
42	Techniques and Representations in Computer Vision	Motion Detection, Optical Flow, and Tracking	Lecture	DS 4150.6	Class Quiz End Term

43	Techniques and Representations in Computer Vision	Local Features and Invariance	Lecture	DS 4150.6	Class Quiz End Term
44	Techniques and Representations in Computer Vision	Bag-of-Words Models, Fisher Vector	Lecture	DS 4150.6	Class Quiz End Term
45	Techniques and Representations in Computer Vision	Object Parts and Attributes	Lecture	DS 4150.6	Class Quiz End Term
46	Techniques and Representations in Computer Vision	Embedding and Representing Objects	Lecture	DS 4150.6	Class Quiz End Term
47	Techniques and Representations in Computer Vision	Understanding and Organizing Parts and Structures	Lecture	DS 4150.6	Class Quiz End Term
48	Techniques and Representations in Computer Vision	Understanding and Organizing Parts and Structures	Lecture	DS 4150.6	Class Quiz End Term

CO	STATEMENT	CORRELATION WITH PROGRAM OUTCOMES (POs)												CORRELATION WITH PROGRAM SPECIFIC OUTCOMES (PSOs)			
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	
[DS 4150.1]	Remember key concepts and terminologies in image processing system.		3		1	1		2	2	2		2	3	1	2		
[DS 4150.2]	Explain pixel connectivity concepts and basic image processing operations.		2		1		1	3	1	1	2		2	2	1	1	
[DS 4150.3]	Apply various image pre-processing techniques to enhance digital images.	3		2	2		3	1	2	2	2			1	3	2	
[DS 4150.4]	Analyse digital images and extract features using different feature extraction techniques.	2	1		2	1	3		1		2	1			2	1	
[DS 4150.5]	Evaluate and justify the use of specific segmentation techniques for different image processing tasks.	2		1	2	1	3			2		1	1	2	3	1	
[DS 4150.6]	Implement solutions for real-world applications using image processing and computer vision techniques.		1	3		2	2		1			2		1	2	3	